

Jounghyun Yoo

Postdoctoral Fellow, Department of Medical Engineering | California Institute of Technology (Caltech)
E-mail: yjh6798@caltech.edu / yjh6798@gmail.com | Phone: +1-626-365-3541 | Website: jounghyunyoo.com

EDUCATION

Pohang University of Science and Technology (POSTECH) Ph.D. in Chemical Engineering Advisor: Prof. Shi-Woo Rhee	Mar. 2014 - Aug. 2019
Pohang University of Science and Technology (POSTECH) B.S. in Chemical Engineering, Magna cum laude	Mar. 2010 - Feb. 2014

PROFESSIONAL EXPERIENCE

California Institute of Technology (Caltech) Postdoctoral Researcher, Department of Medical Engineering, Division of Engineering and Applied Science Advisor: Prof. Wei Gao	Sep. 2021 - Present
Ulsan National Institute of Science and Technology (UNIST) Postdoctoral Researcher, Department of Biomedical Engineering Advisor: Prof. Jinmyoung Joo	Sep. 2021 - Sep. 2022; Jan. 2024 – Aug. 2026
Korea Institute of Science and Technology (KIST) Postdoctoral Researcher, Center for Theragnosis, Biomedical Research Institute Advisor: Dr. Sehoon Kim	Sep. 2019 - Aug. 2021

RESEARCH INTERESTS

Medical micro/nanorobotics. Designing micro/nanorobotic systems for active biological transport, barrier penetration, targeted localization, controlled therapeutic delivery, and adaptive interaction with biological microenvironments.

Imaging-guided biofabrication. Developing non-invasive biofabrication strategies to construct functional biomaterial interfaces within living tissues for tissue-integrated monitoring, localized intervention, and regenerative medicine.

Bioactive nanomaterials. Engineering biocompatible and biodegradable nanomaterial platforms with tunable physicochemical properties and biological functions to regulate cellular interactions, biodistribution, imaging contrast, and therapeutic performance.

SELECTED PUBLICATIONS

†Equal contribution; *Corresponding author.

1. **J. Yoo**†, S. Kim†, W. Yoo, H. Han, G. Kim, A. W. Tessema, S. Kim, S. Tang, J. Kim, H. D. Truong, X. Ma, G. Kim, S. Won, J.-W. Choi, J. Hwang, J. Oh, Y. Seok, J. K. Kim, H. Cho, T.-E. Park, J. Joo, W. Gao, Photothermally gated nanorobots for active blood-brain barrier transcytosis and brain delivery. *Science Robotics*, under revision.
2. H. Han†, X. Ma†, **J. Yoo**†, S. Tang, K. Zhao, E. Criado-Hidalgo, H. D. Truong, C. Fezzi, J. Zhang, G. Chen, C. Hays, R. J. Reyes Segura, Z. Jin, A. Liu, M. G. Shapiro, J. R. Greer, Q. Zhou, W. Gao, Programmable multigas cancer therapy using ultrasound-propelled microrobots. *Nature Communications*, under revision.
3. **J. Yoo**, J. Joo, W. Gao, Bridging the barrier: Micro- and nanorobotics for active brain delivery. *Matter*, in press (2026). doi: 10.1016/j.matt.2026.102858.
4. M. Kim, K. Kim, J. Lee, S. A. Park, E. Jeong, S.-Y. Choi, T. Kwon, K. Myung, **J. Yoo***, S. W. Cho*, J. Joo*, Directed and rapid intracellular delivery of CRISPR/Cas9 ribonucleoproteins using charge-reversible fusogenic nanoparticles for therapeutic genome editing. *Small*, 22, e11362 (2026). Featured on Journal Cover.
5. S. Lee†, **J. Yoo**†, G. Bae, R. Thangam, J. Heo, J. Y. Park, H. Choi, C. Kim, J. An, J. Kim, K. R. Mun, S. Shin, K. Zhang, P. Zhao, Y. Kim, N. Kang, S.-B. Han, D. Kim, J. Yoon, M. Kang, J. Kim, L. Yang, S. Karamikamkar, J. Kim, Y. Zhu, A. H. Najafabadi, G. Song, D.-H. Kim, K.-B. Lee, S. J. Oh, H.-D. Jung, H.-C. Song, W. Y. Jang, L. Bian, Z. Chu, J. Yoon, J. S. Kim, Y. S. Zhang, Y. Kim, H. S. Jang, S. Kim, H. Kang, Photonic control of ligand nanospacing in self-assembly regulates stem cell fate. *Bioactive Materials*, 34, 164 (2024).
6. **J. Yoo**†, S. Tang†, W. Gao, Micro- and nanorobots for biomedical applications in the brain. *Nature Reviews Bioengineering*, 1, 308 (2023).

PUBLICATIONS

†Equal contribution; *Corresponding author.

1. **J. Yoo**†, S. Kim†, W. Yoo, H. Han, G. Kim, A. W. Tessema, S. Kim, S. Tang, J. Kim, H. D. Truong, X. Ma, G. Kim, S. Won, J.-W. Choi, J. Hwang, J. Oh, Y. Seok, J. K. Kim, H. Cho, T.-E. Park, J. Joo, W. Gao, Photothermally gated nanorobots for active blood-brain barrier transcytosis and brain delivery. *Science Robotics*, under revision.
2. H. Han†, X. Ma†, **J. Yoo**†, S. Tang, K. Zhao, E. Criado-Hidalgo, H. D. Truong, C. Fezzi, J. Zhang, G. Chen, C. Hays, R. J. Reyes Segura, Z. Jin, A. Liu, M. G. Shapiro, J. R. Greer, Q. Zhou, W. Gao, Programmable multigas cancer therapy using ultrasound-propelled microrobots. *Nature Communications*, under revision.
3. **J. Yoo**, J. Joo, W. Gao, Bridging the barrier: Micro- and nanorobotics for active brain delivery. *Matter*, in press, doi: 10.1016/j.matt.2026.102858.
4. S. Tang, H. Han, X. Ma, P. N. Patel, C. Gong, J. Zhang, E. Criado-Hidalgo, **J. Yoo**, J. Li, G. Kim, M. G. Shapiro, Q. Zhou, W. Gao, Enzymatic microbubble robots. *Nature Nanotechnology*, 21, 397 (2026).
5. M. Kim, K. Kim, J. Lee, S. A. Park, E. Jeong, S.-Y. Choi, T. Kwon, K. Myung, **J. Yoo***, S. W. Cho*, J. Joo*, Directed and rapid intracellular delivery of CRISPR/Cas9 ribonucleoproteins using charge-reversible fusogenic nanoparticles for therapeutic genome editing. *Small*, 22, e11362 (2026). Featured on Journal Cover.
6. **J. Yoo**†, Y. Kim†, J. H. Back†, J. Shin, P. K. Bae, K. M. Park, M. Kim, Y. H. Seo, Y. Bak, Y. H. Heo, J. Heo, H. Choi, Y. Kim, S. Lee, J. E. Lee, S. Jeong, J.-K. Yang, S. Kim, Surface-engineered nanobeads for regiospecific antibody binding: A robust immunoassay platform leveraging catalytic signal amplification. *Biosensors and Bioelectronics*, 281, 117463 (2025).
7. E. Davoodi, J. Li, X. Ma, A. H. Najafabadi, **J. Yoo**, G. Lu, E. S. Sani, S. Lee, H. Montazerian, G. Kim, J. W. Yang, J. Williams, Y. Zeng, L. S. Li, Z. Jin, B. Sadri, S. S. Nia, L. V. Wang, T. K. Hsiai, P. S. Weiss, Q. Zhou, A. Khademhosseini, D. Wu, M. G. Shapiro, W. Gao, Imaging-guided deep tissue in vivo sound printing. *Science*, 388, 616 (2025).
8. Y. Choi, P. Jin, S. Lee, Y. Song, R. Y. Tay, G. Kim, **J. Yoo**, H. Han, J. Yeom, J. H. Cho, D. Kim, W. Gao, All-printed chip-less wearable neuromorphic system for multimodal physicochemical health monitoring. *Nature Communications*, 16, 5689 (2025).
9. J. Seong, M. Kim, **J. Yoo**, D. L. Mack, J. H. Lee, J. Joo, Sustained release of HIF-2 α inhibitors using biodegradable porous silicon carriers for enhanced immunogenic cell death of malignant Merkel cell carcinoma. *ACS Applied Materials & Interfaces*, 17, 7449 (2025).
10. S. Lee†, **J. Yoo**†, G. Bae, R. Thangam, J. Heo, J. Y. Park, H. Choi, C. Kim, J. An, J. Kim, K. R. Mun, S. Shin, K. Zhang, P. Zhao, Y. Kim, N. Kang, S.-B. Han, D. Kim, J. Yoon, M. Kang, J. Kim, L. Yang, S. Karamikamkar, J. Kim, Y. Zhu, A. H. Najafabadi, G. Song, D.-H. Kim, K.-B. Lee, S. J. Oh, H.-D. Jung, H.-C. Song, W. Y. Jang, L. Bian, Z. Chu, J. Yoon, J. S. Kim, Y. S. Zhang, Y. Kim, H. S. Jang, S. Kim, H. Kang, Photonic control of ligand nanospacing in self-assembly regulates stem cell fate. *Bioactive Materials*, 34, 164 (2024).
11. H. Han, X. Ma, W. Deng, J. Zhang, S. Tang, O. S. Pak, L. Zhu, E. Criado-Hidalgo, C. Gong, E. Karshalev, **J. Yoo**, M. You, A. Liu, C. Wang, H. Shen, P. N. Patel, C. L. Hays, P. J. Gunnarson, L. Li, Y. Zhang, J. O. Dabiri, L. V. Wang, M. G. Shapiro, D. Wu, Q. Zhou, J. R. Greer, W. Gao, Imaging-guided bioresorbable acoustic hydrogel microrobots. *Science Robotics*, 9, eadp3593 (2024).
12. S. Tang, D. Tang, H. Zhou, Y. Li, D. Zhou, X. Peng, C. Ren, Y. Su, S. Zhang, H. Zheng, F. Wan, **J. Yoo**, H. Han, X. Ma, W. Gao, S. Wu, Bacterial outer membrane vesicle nanorobot. *Proceedings of the National Academy of Sciences*, 121, e2403460121 (2024).
13. H. Chang, Y. Park, K. Kim, C. Han, Y. Yoon, W. Yoo, **J. Yoo**, D. Lee, H. Han, K. Kim, J. Joo, W. Kwon, Room-temperature phosphorescence of defect-engineered silica nanoparticles for high-contrast afterglow bioimaging. *Chemical Engineering Journal*, 493, 152529 (2024).
14. **J. Yoo**†, S. Tang†, W. Gao, Micro- and nanorobots for biomedical applications in the brain. *Nature Reviews Bioengineering*, 1, 308 (2023).
15. **J. Yoo**†, K. Kim†, S. Kim, H. H. Park, H. Shin, J. Joo, Tailored polyethylene glycol grafting on porous nanoparticles for enhanced targeting and intracellular siRNA delivery. *Nanoscale*, 14, 14482 (2022). Featured on Journal Cover.
16. T. F. Anjong†, H. Choi†, **J. Yoo**†, Y. Bak, Y. Cho, D. Kim, S. Lee, K. Lee, B.-G. Kim, S. Kim, Multifunction-harnessed afterglow nanosensor for molecular imaging of acute kidney injury in vivo. *Small*, 18, 2200245 (2022).
17. Y.-D. Lee†, H. J. Shin†, **J. Yoo**†, G. Kim, M.-K. Kang, J. J. Lee, J. Bang, J.-K. Yang, S. Kim, Metal complexation-mediated stable and biocompatible nanoformulation of clinically approved near-infrared absorber for improved tumor targeting and photonic theranostics. *Nano Convergence*, 8, 1 (2021).
18. Y. Kim, R. Thangam, **J. Yoo**, J. Heo, J. Y. Park, N. Kang, S. Lee, J. Yoon, K. R. Mun, M. Kang, S. Min, S. Y. Kim, S. Son, J. Kim, H. Hong, G. Bae, K. Kim, S. Lee, L. Yang, J. Y. Lee, J. Kim, S. Park, D.-H. Kim, K.-B. Lee, W. Y. Jang, B. H.

- Kim, R. Paulmurugan, S.-W. Cho, H.-C. Song, S. J. Kang, W. Sun, Y. Zhu, J. Lee, H.-J. Kim, H. S. Jang, J. S. Kim, A. Khademhosseini, Y. Kim, S. Kim, H. Kang, Photoswitchable microgels for dynamic macrophage modulation. *Advanced Materials*, 34, e2205498 (2022). Featured on Journal Cover.
19. D. Jang, S. Yu, K. Chung, **J. Yoo**, F. M. Mota, J. Wang, D. J. Ahn, S. Kim, D. H. Kim, Direct deposition of anatase TiO₂ on thermally unstable gold nanobipyramid: Morphology-conserved plasmonic nanohybrid for combinational photothermal and photocatalytic cancer therapy. *Applied Materials Today*, 27, 101472 (2022).
 20. K. Jeong, D. Kim, H. J. Kim, Y.-D. Lee, **J. Yoo**, D. Jang, S. Lee, H. Park, Y. Kim, A. Singh, D. J. Ahn, D. H. Kim, J. Bang, J. Kim, P. N. Prasad, S. Kim, Photoecheogenic inflatable nanohybrids for upconversion-mediated sonotheranostics. *ACS Nano*, 15, 18394 (2021).
 21. J. Shin, D. M. Kang, **J. Yoo**, J. Heo, K. Jeong, J. H. Chung, Y. S. Han, S. Kim, Superoxide-responsive fluorogenic molecular probes for optical bioimaging of neurodegenerative events in Alzheimer's disease. *Analyst*, 146, 4748 (2021). Featured on Journal Cover.
 22. **J. Yoo**†, H. Kim†, H. Chang†, W. Park, S. K. Hahn, W. Kwon, Biocompatible organosilica nanoparticles with self-encapsulated phenyl motifs for effective UV protection. *ACS Applied Materials & Interfaces*, 12, 9062 (2020).
 23. **J. Yoo**†, U. Ryu†, W. Kwon, K. M. Choi, A multi-dye containing MOF for the ratiometric detection and simultaneous removal of Cr₂O₇²⁻ in the presence of interfering ions. *Sensors and Actuators B: Chemical*, 283, 426 (2019).
 24. Y. Park†, **J. Yoo**†, M.-H. Kang*, W. Kwon, J. Joo, Photoluminescent and biodegradable porous silicon nanoparticles for biomedical imaging. *Journal of Materials Chemistry B*, 7, 6271 (2019). Featured on Journal Cover.
 25. H. Chang†, **J. Yoo**†, H. Kim, W. Park, S. K. Hahn, W. Kwon, Controlled growth of silica nanoparticles using two-phase orthogonal solvents for bioimaging. *Journal of Luminescence*, 214, 116529 (2019).
 26. T. Lee, H. T. Bui, **J. Yoo**, M. Ra, S. H. Han, W. Kim, W. Kwon, Formation of TiO₂@carbon core/shell nanocomposites from single molecular layer of aromatic compounds for photocatalytic hydrogen peroxide generation. *ACS Applied Materials & Interfaces*, 11, 41196 (2019).
 27. **J. Yoo**†, S. Han†, W. Park, T. Lee, Y. Park, H. Chang, S. K. Hahn, W. Kwon, Defect-induced fluorescence of silica nanoparticles for bioimaging applications. *ACS Applied Materials & Interfaces*, 10, 44247 (2018).
 28. D. Lee†, S. Beack†, **J. Yoo**†, S.-K. Kim, C. Lee, W. Kwon, S. K. Hahn, C. Kim, In vivo photoacoustic imaging of livers using biodegradable hyaluronic acid-conjugated silica nanoparticles. *Advanced Functional Materials*, 28, 1800941 (2018). Featured on Journal Cover.
 29. U. Ryu†, **J. Yoo**†, W. Kwon, K. M. Choi, Tailoring nanocrystalline metal-organic frameworks as fluorescent dye carriers for bioimaging. *Inorganic Chemistry*, 56, 12859 (2017).
 30. Y. Park†, **J. Yoo**†, B. Lim, W. Kwon, S.-W. Rhee, Improving functionality of carbon nanodots: Doping and surface functionalization. *Journal of Materials Chemistry A*, 4, 11582 (2016).

MANUSCRIPTS IN PREPARATION

1. **J. Yoo**, D.-M. Lee, Y. E. Cho, W. Gao, Imaging-guided in vivo biofabrication for dynamic tissue interfacing. In preparation.

PATENTS

1. H. Kang, S. Kim, H. S. Jang, Y. Kim, **J. Yoo**, J. Heo, G. Bae, N. Kang, S. Lee, Y. Kim, Self-assembled complex controlled by light energy for stem cells and nanosystem including same, Korean patent (application: 10-2023-0029829, 7 Mar. 2023; granted: 10-2897463-0000, 3 Dec. 2025).
2. W. Kwon, **J. Yoo**, S. K. Hahn, H. Kim, Ultraviolet protection silica nanoparticle composition and its manufacturing method, Korean patent (application: 10-2019-0173419, 23 Dec. 2020; granted: 10-2340304-0000, 13 Dec. 2021).
3. K. M. Choi, W. Kwon, U. Ryu, **J. Yoo**, Imaging-agents loaded nanocrystalline metal-organic frameworks for containing and method for preparing the same, Korean patent (application: 10-2017-0052355, 24 Apr. 2017; granted: 10-1890588-0000, 16 Aug. 2018).

HONORS AND FELLOWSHIPS

UNIST Global Postdoctoral Short-term Training Fellowship	Sep. 2025 - Dec. 2025
Overseas postdoctoral research fellowship supported by Ulsan National Institute of Science and Technology (UNIST)	
Sejong Science Fellowship (Overseas Training Track)	Sep. 2024 - Aug. 2025
Overseas postdoctoral research fellowship supported by the Ministry of Science and ICT (MSIT), Republic of Korea	
Korea Health Industry Development Institute (KHIDI) Fellowship	Sep. 2021 - Sep. 2022
Overseas postdoctoral research fellowship supported by the Ministry of Health and Welfare, Republic of Korea	

Best Paper of the Year
Awarded by the Department of Chemical Engineering at POSTECH
National Scholarship for Science and Engineering
Supported by Korea Student Aid Foundation (KOSAF)

Nov. 2018

Mar. 2010 - Feb. 2014

TEACHING, MENTORING, AND LEADERSHIP

Postdoctoral Researcher Coordinator, Caltech Appointed by the Caltech Korean Graduate Student Association (CKGSA)	Jan. 2024 - Sep. 2024
Project Leader and Research Mentor, KIST Mentored undergraduate and doctoral researchers in bioactive nanomaterials for biomedical applications	Fall 2019 - Spring 2021
Teaching Assistant, POSTECH Solid State Chemical Engineering (CHEB360) Physical Chemical Laboratory (CHEB212) Introduction to Chemical Engineering (CHEB100)	Fall 2014 - Spring 2018
Peer Tutor, POSTECH Chemical Engineering Thermodynamics (CHEB204)	Spring 2013
Student Advisor, POSTECH Future Planning for College Life I (MSUS102)	Spring 2012

REFERENCES

Prof. Wei Gao
Professor of Medical Engineering
California Institute of Technology (Caltech)
Email: weigao@caltech.edu

Prof. Jinmyoung Joo
Associate Professor of Biomedical Engineering
Ulsan National Institute of Science and Technology (UNIST)
Email: jjoo@unist.ac.kr

Dr. Sehoon Kim
Principal Research Scientist
Chemical and Biological Integrative Research Center
Korea Institute of Science and Technology (KIST)
Email: sehoonkim@kist.re.kr

Prof. Shi-Woo Rhee
Professor Emeritus of Chemical Engineering
Pohang University of Science and Technology (POSTECH)
Email: srhee@postech.ac.kr